

Improve the experience of self- propelled wheelchair users

By: Zi Qian, Jing Heng, Jerry,
Isaiah



What is the problem

Isaiah



Wheelchair user having a hard time moving up the slope

How might we create a supportive solution for wheelchair users that makes going up and down slopes easier, safer, and less tiring while reducing the risk of falls?

Ideation process - Crazy 8

Motor - Assist Switch wheelchair

1



Auto brake system

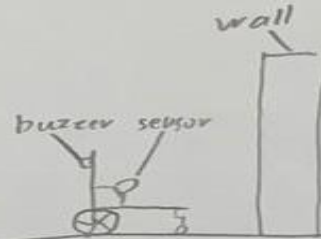
2

aut. brake



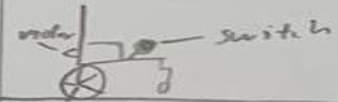
Ultrasonic obstacle Buzzer

3



Potentiometer speed control (user can adjust speed manually)

4



Tilt Sensor Emergency Stop

5



Remote - controlled Assist

6

remote - controlled Assist using phone



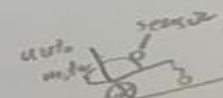
Anti roll back system

7



Smart slope detection system

8



Market Study - Value effort map

Isaiah

Value for users

High value, high effort	High value, low effort
• 5 Tilt Sensor Emergency Stop → Stops wheelchair if it tilts dangerously. • 8 Smart Slope Detection System → Uses angle sensors to auto -adjust motor power. • 2 Auto Brake System → Automatically brakes when going downhill too fast.	• 1 Motor assist switch wheelchair → Simple switch lets motor help push up slopes. • 4 Potentiometer Speed Control → User controls how fast the motor run
Low value, high effort	Low value, Low effort
• 6 Remote-Controlled Assist → Needs remote + receiver, not practical for user independence. • 7 Anti-Rollback System → Great idea but hard to build without mechanical parts.	• 3 Ultrasonic Obstacle Buzzer → Alerts when near obstacle, but doesn't assist on slope.

Effort by team

Ideation process

- First – we brainstorm our idea using crazy 8
- Second – we decided that we choose Motor-assist switch wheelchair , with ultrasonic sensor and potentiometer speed control
 - Reason - it is high in value low in effort

Market Study



Power assist devices



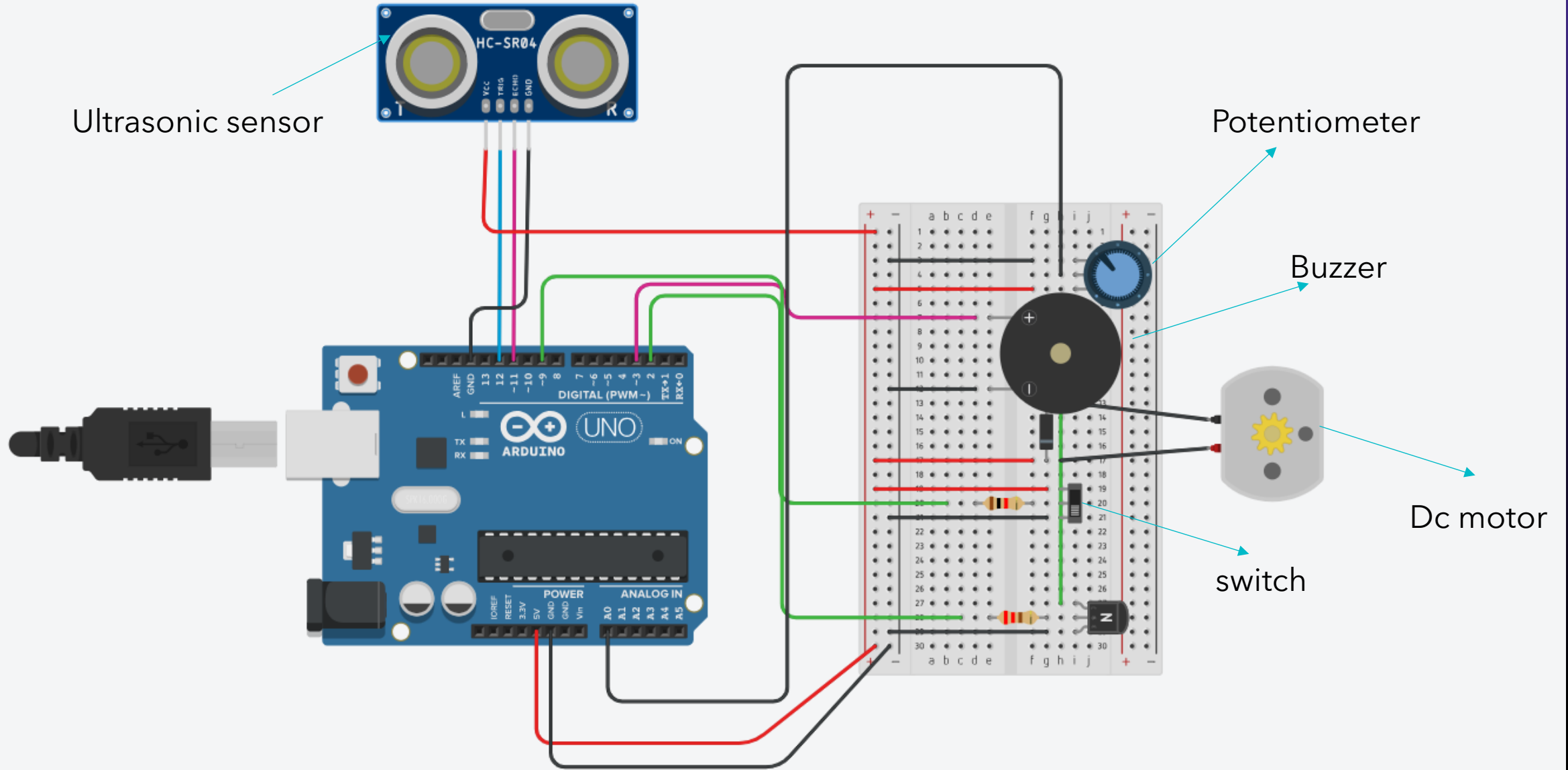
Free wheel attachment

Market Study –key Differentiating factor

Differences	Power Assist Devices	Free Wheel Attachment	My idea
Speed adjustment	 Yes	 No	 Yes
Weight	Light	Heavy	Light
Cost	\$6,626	\$599	\$600 assume

Features	Power assist Devices	Free Wheel attachment	Our Idea
Alert if got danger	 No	 No	 Yes
Make it easier to go up and down the slope	 Yes	 Yes	 Yes
User Effort Reduction	 Yes	 No	 Yes

System & Circuit design

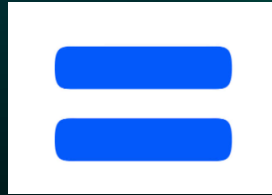


Explain which is the input and which is the output

Input



Ultrasonic Sensor

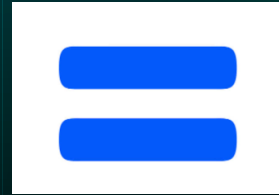


Output



Piezo Buzzer

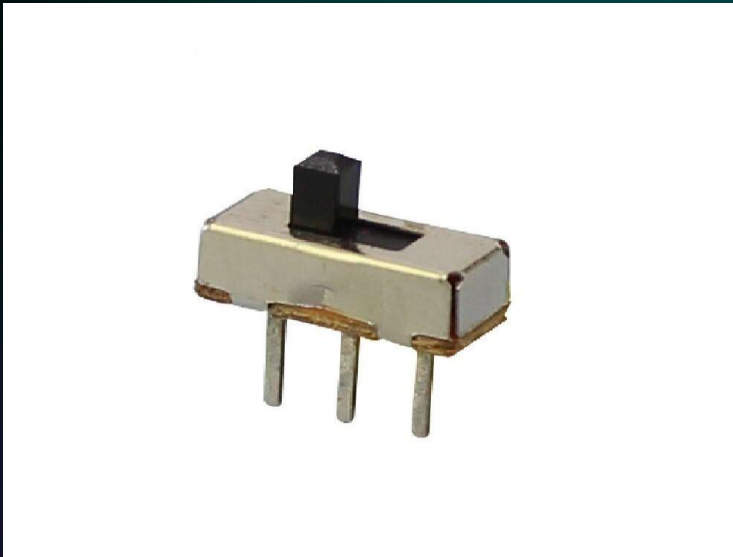
Input



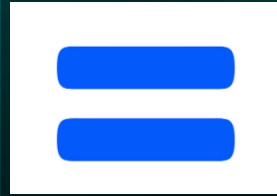
Output



Input



Switch

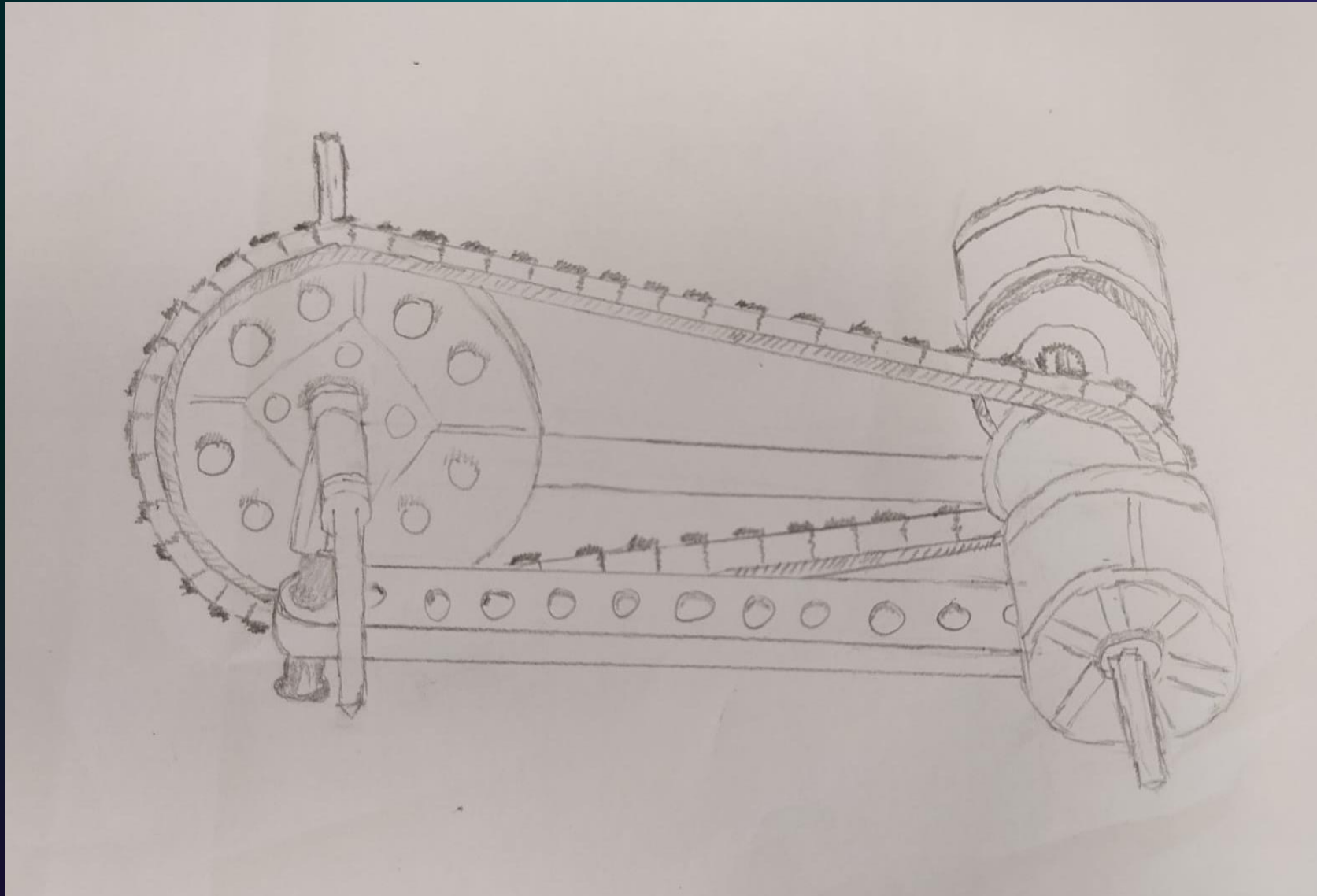


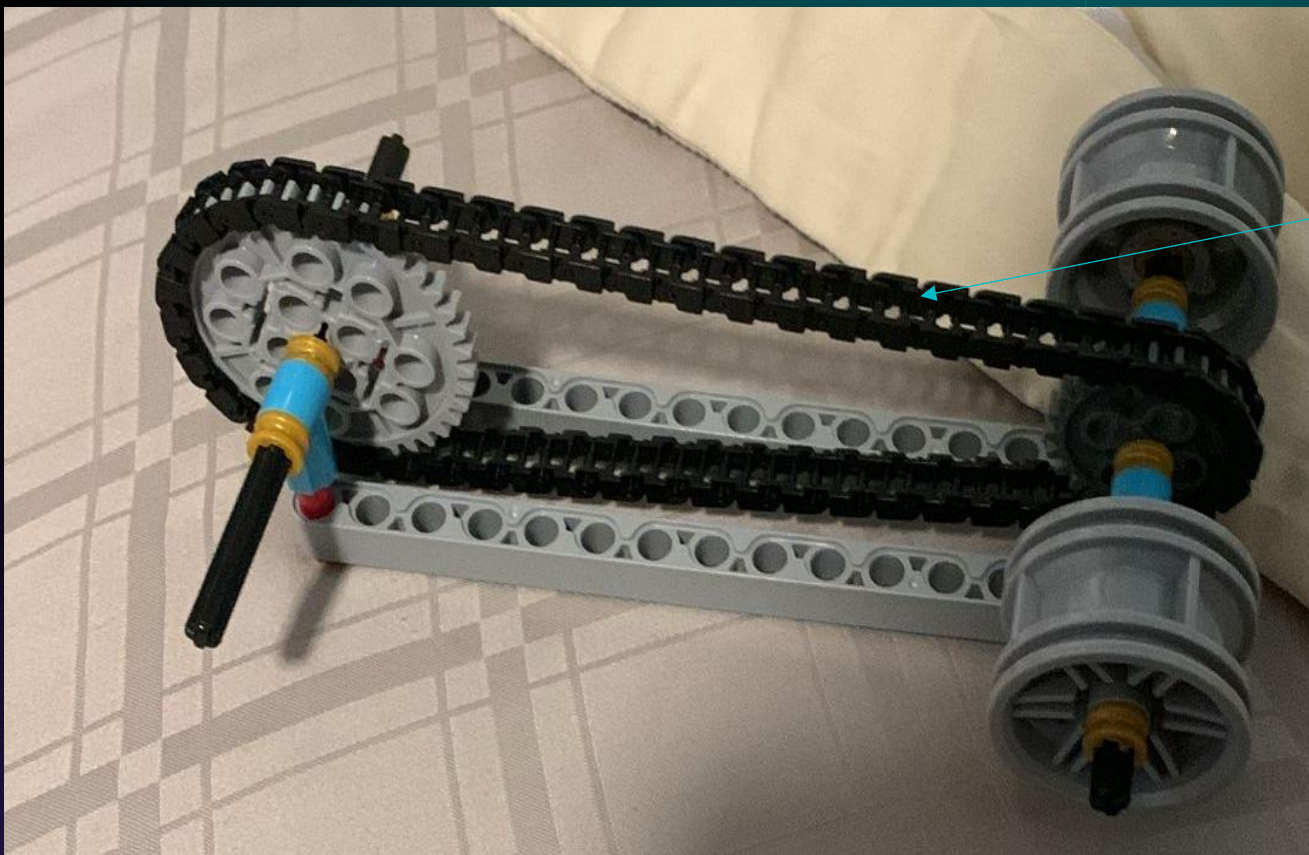
Output



Mini Chain-Driven Transmission System

Jingheng





Why use chain?

We will put the
hardware over
here



Jingheng



We will put the potentiometer , the switch buzzer here

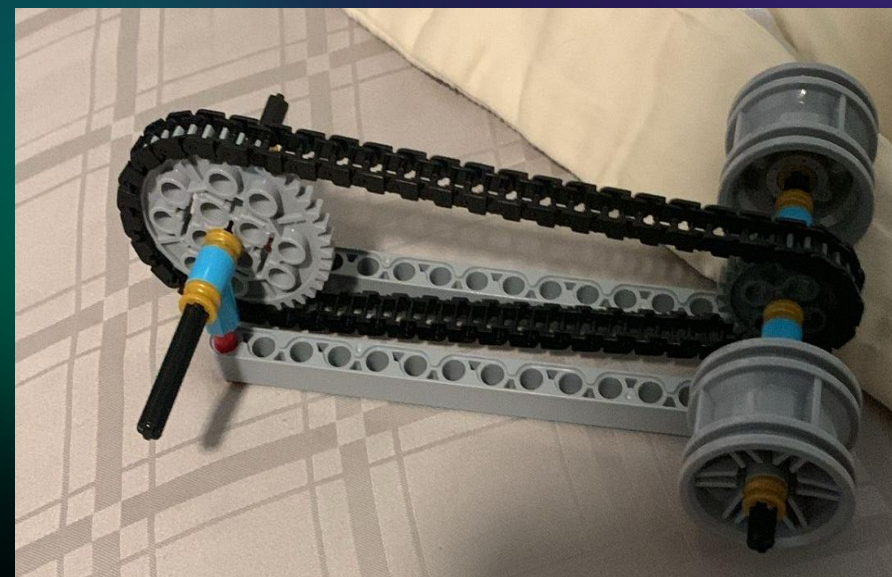
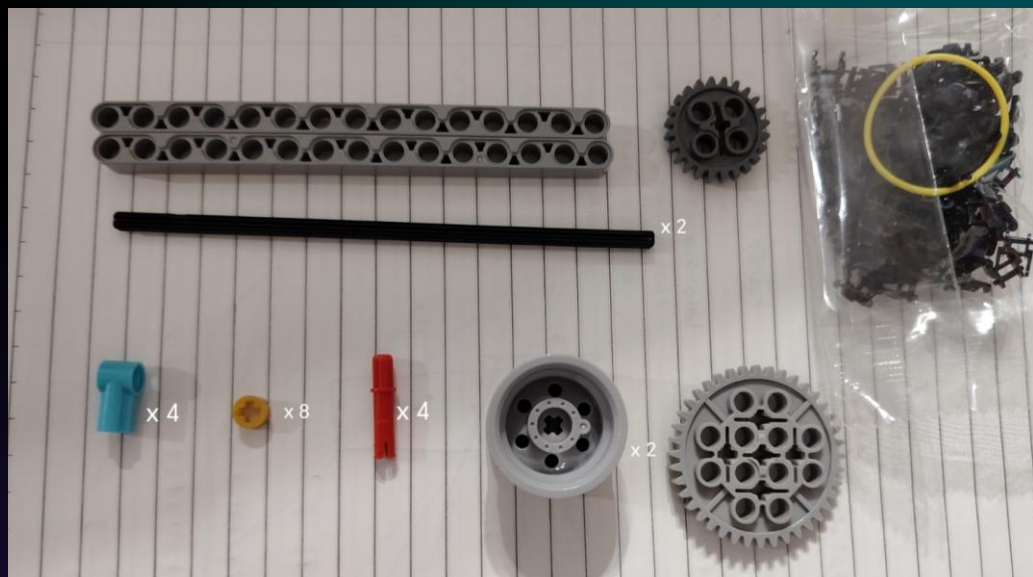
We put the ultrasonic sensor to detect the obstacle

Explain how this would help the wheelchair user?

- It has a dc motor that help to turn the wheel forward
- It has a ultrasonic sensor to detects obstacles in front of the wheelchair. If something is too close, it can warn the user or even stop the wheelchair to prevent accidents like tipping or falling.
- It have potentiometer which act like the speed controller so user can turn it to increase or decrease the motor speed, depending on their comfort or the environment.

PROTOTYPE DEVELOPMENT

Hardware prototype development

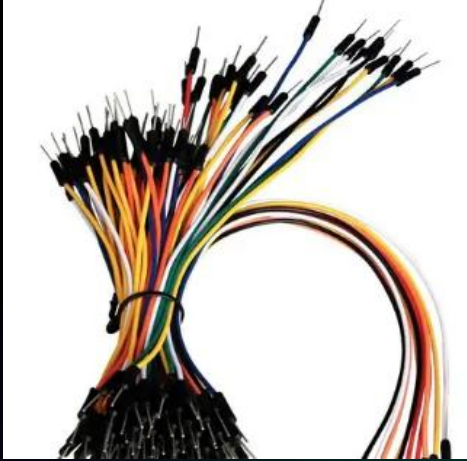


Software

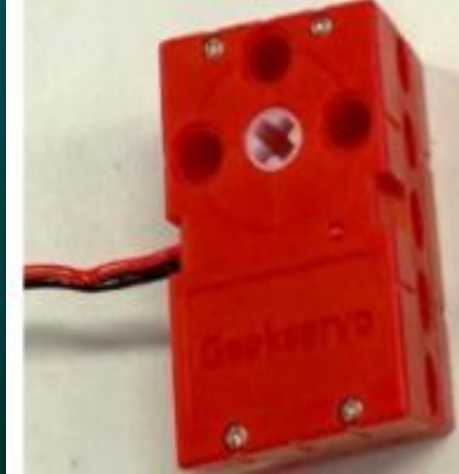
Component

Jingheng

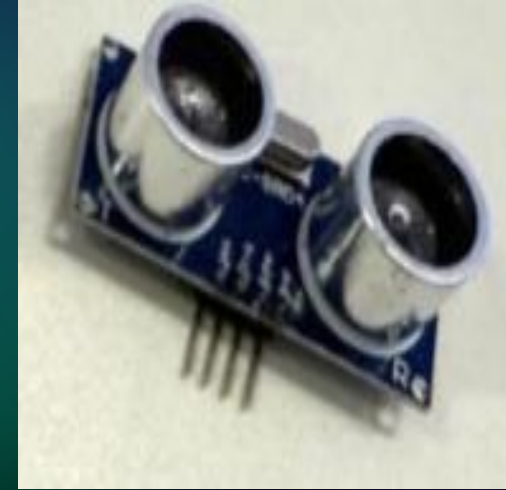
Wire



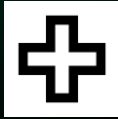
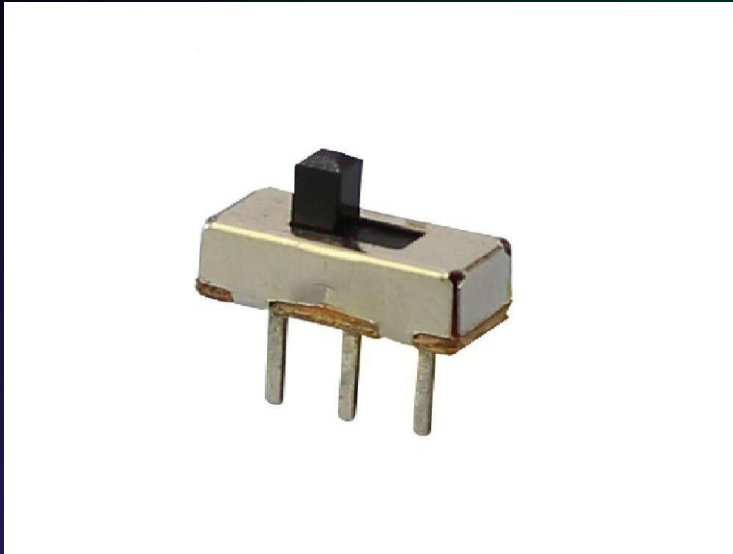
Dc motor



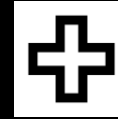
Ultra sonic sensor



Switch

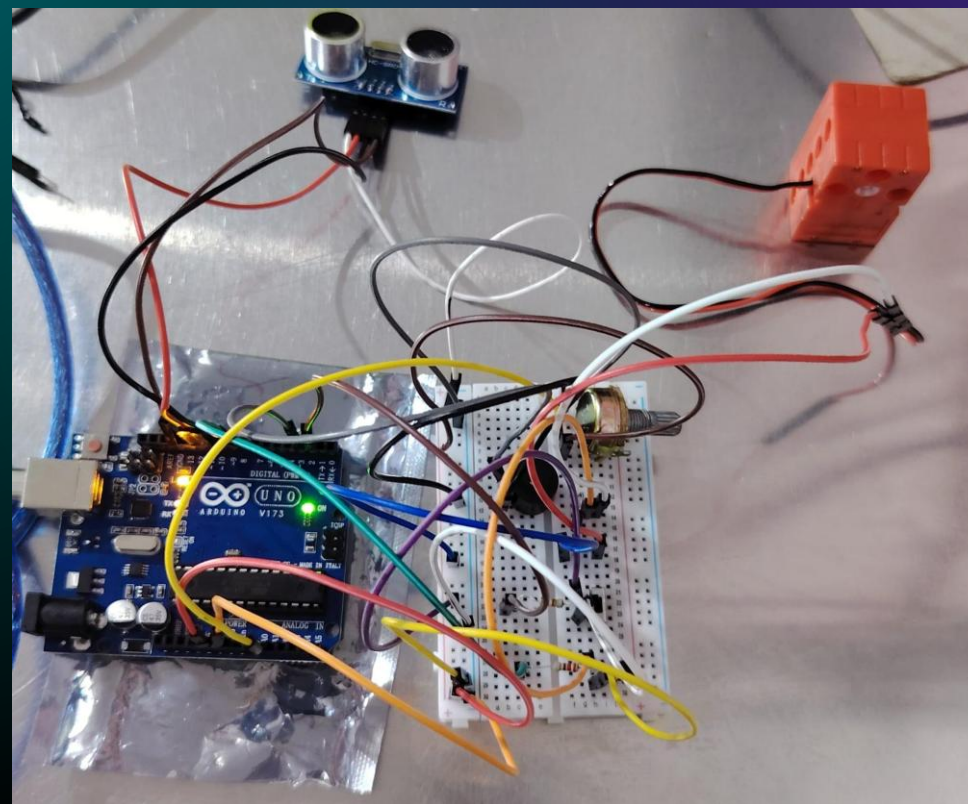
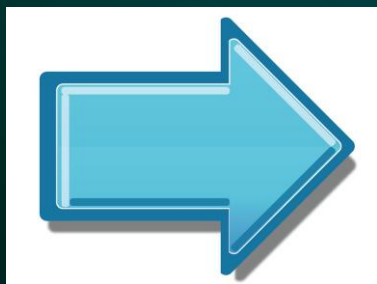


Potentiometer



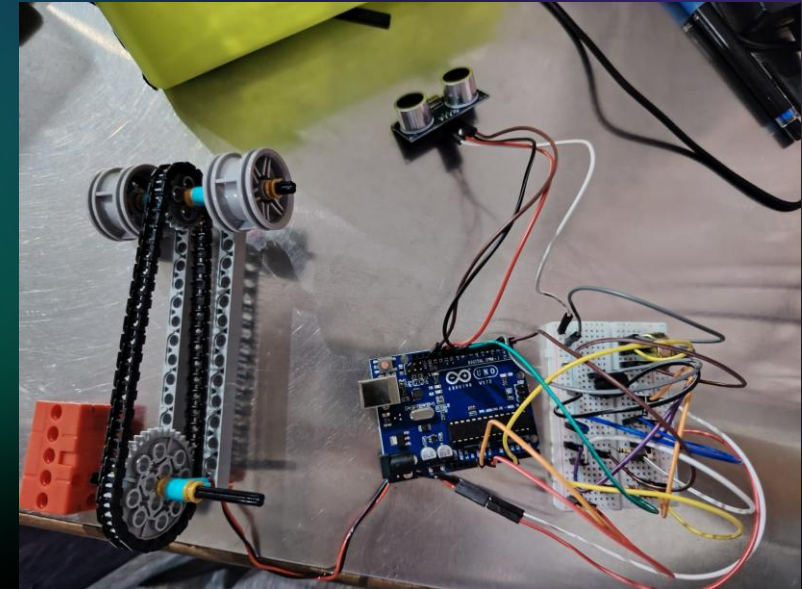
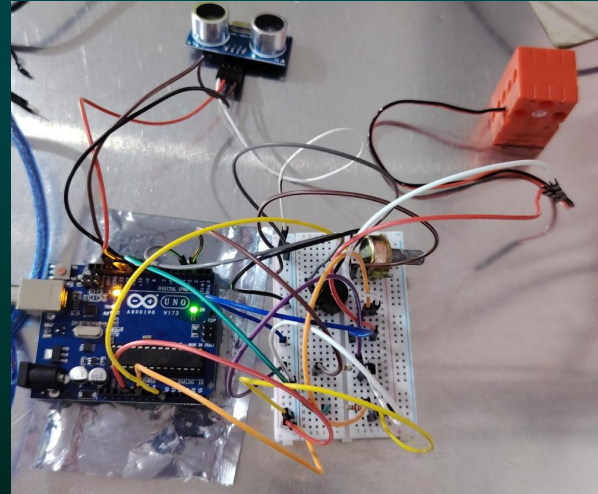
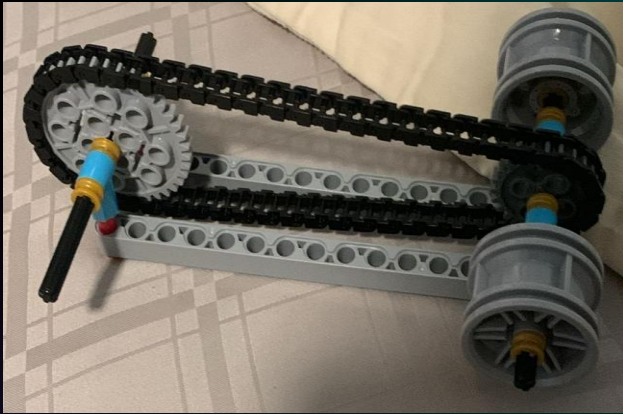
Buzzer



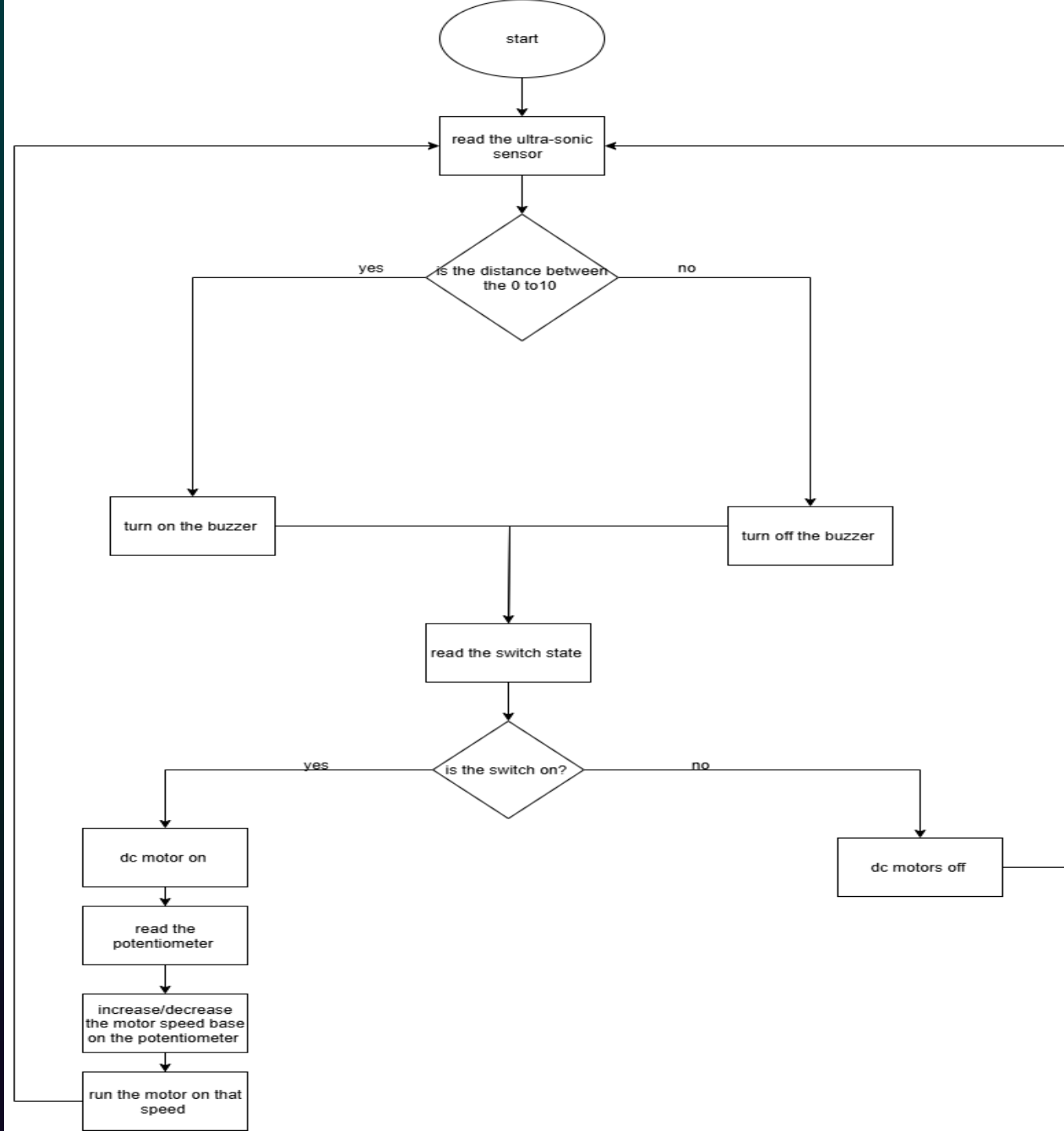


Hardware + software

Jingheng



What is this system about?



TESTING AND REFINEMENT

Component	Button	Switch	Led	Buzzer	Ultrasonic sensor	Potentiometer
Design 1						
Design 2			 (remove)			
Design 3 (Final Design)	 (replace)					

At first ultrasonic placement

ziqian

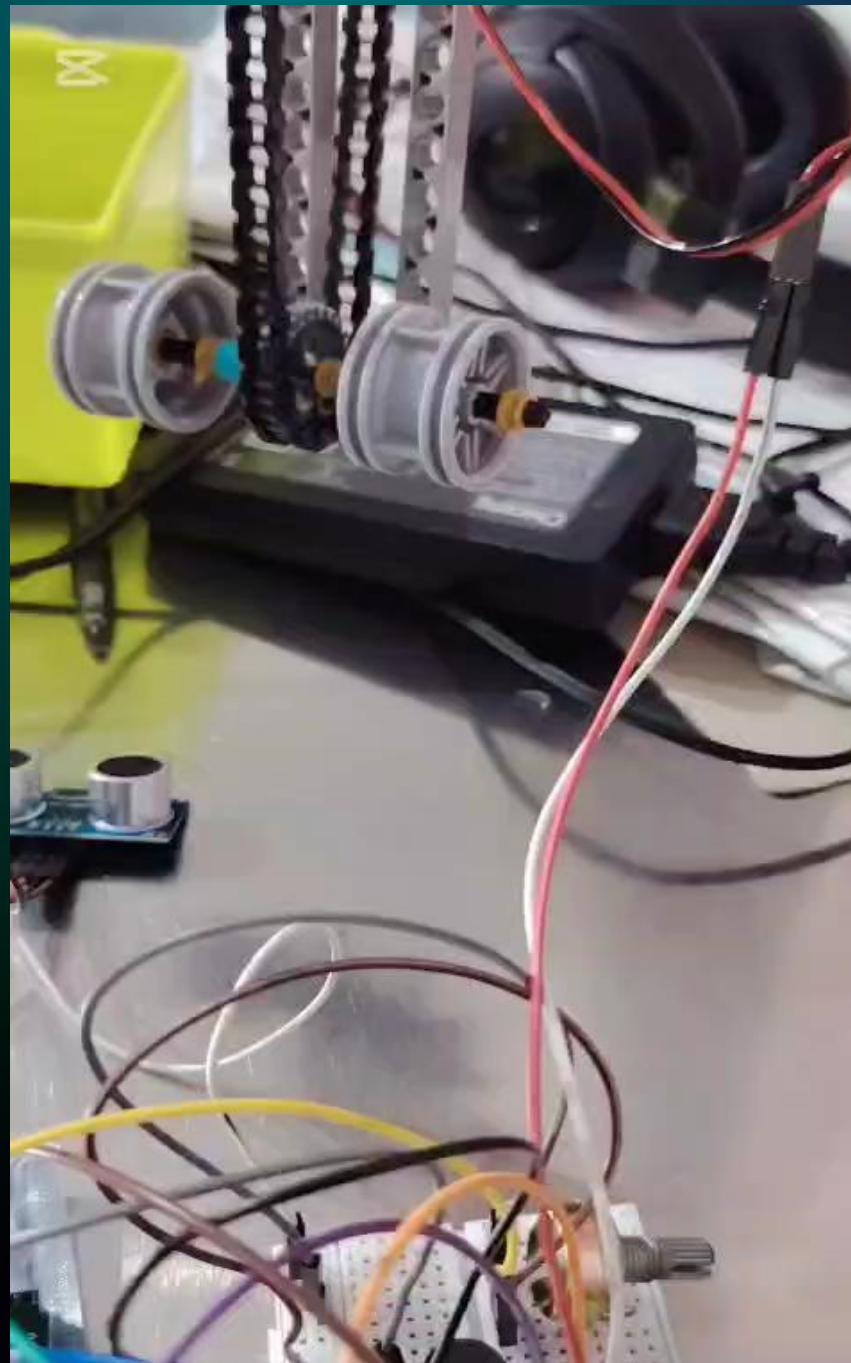


Now ultrasonic placement

ziquan



Challenges faced during the project	Solution
Building to the circuit	Look at the lectures note to learn how to build The circuit
Coding the Arduino	Using YouTube video and lectures note
Building the hardware	We look at the practice that we did during the class then we implement it to our hardware



ziqian

Conclusion

ziquan

- This project help to reduce the effort of the wheelchair user
- Helps reduce risk of tipping or collision.
- Easy to use and affordable for everyone
- Improve can make
 - we use a better Dc motor, power supply and potentiometer

Q&A

Citation

https://www.amazon.com/FreeWheel-Wheelchair-Attachment-Pneumatic-Footrests/dp/B06ZYW98HW?utm_source

https://bestpowerwheelchair.com/products/smartdrive-mx2?utm_source